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Visible Light-Mediated Coupling of α-Bromochalcones with Alkenes

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Abstract: Photoredox catalyzed intermolecular couplings of α-bromochalcones to olefins have been developed. Employing 1 mol% of the iridium complex $Ir(ppy)_3$ as photocatalyst, vinyl radicals are generated from α-bromochalcones as the key intermediate, which efficiently engage in a formal [4+2] cyclization with various alkenes. The resulting 3,4-dihydronaphthalenes can be readily transformed to the corresponding naphthalenes and further cyclized to 5*H*-benzo[*c*]fluorenes. Alternatively, Heck-type coupling products are obtained with sterically more hindered alkenes or allylated products if the alkene possesses a suitable leaving group in the allylic position.

Keywords: cyclization; dihydronaphthalenes; olefins; vinyl radicals; visible light photocatalysis

Introduction

The dihydronaphthalene ring system is a ubiquitous structural motif in various therapeutically relevant natural products.[1] Such compounds have been used as building blocks in the synthesis of biologically active compounds[2] and of natural products[3] and applied as fluorescent ligands for the estrogen receptors,[4] or as potent aldosterone synthase (CYP11B2) inhibitors for the treatment of congestive heart failure and myocardial fibrosis.[5]

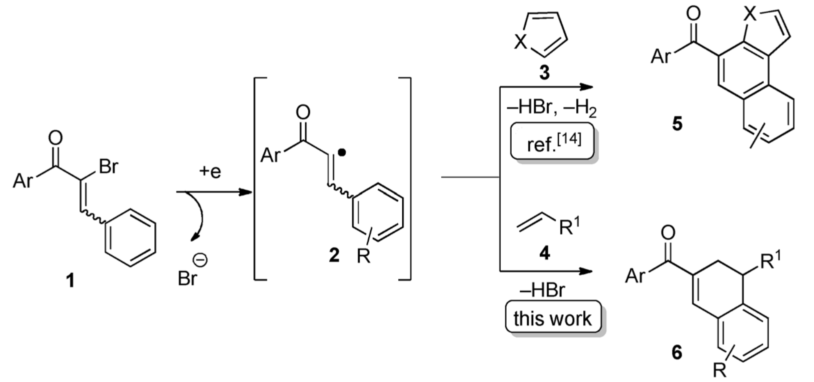
Consequently, considerable efforts have been made for the synthesis of dihydronaphthalenes. One of the most useful method is the dearomatization of naphthalenes by nucleophilic addition of organometallic reagents.[6] Transition metal-catalyzed processes, such as palladium-catalyzed coupling of Grignard reagents with *in-situ* generated enol phosphates,[7] gold(I)-catalyzed intramolecular vinylidenecyclopropane rearrangements,[8] manganese(III)-mediated oxidations of diethyl α-benzylmalonate in the presence of alkynes,[9] copper(II)-catalyzed [4+2] cycloadditions of *ortho*-alkynyl(oxo)benzenes with alkenes,[10] nickel-catalyzed [2+2+2] cycloadditions of arynes, alkenes, and alkynes[11] have also provided efficient accesses to dihydronaphthalenes.

Over the past decade, visible light photocatalysis has been established as a powerful and versatile tool for organic synthesis.[12] As a continuation of the very few existing studies on visible light-promoted generation of vinyl radicals,[13] we have recently demonstrated their utilization in a cascade cyclization between α-bromochalcones **1** and heteroarenes **3** (Scheme 1, *top*), which was accompanied by a dehydrogenation to give rise to polycyclic heteroarenes **5**.[14]

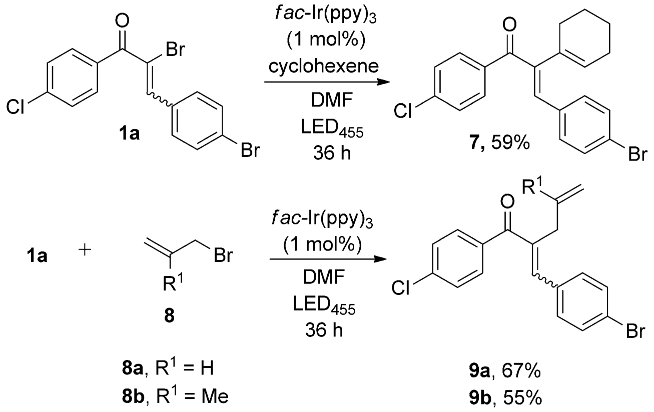
Considering their high reactivity and electrophilic nature, we questioned if vinyl radicals **2** could also engage in a coupling with alkenes, and we report here that a wide variety of terminal alkenes **4** can be effectively coupled to dihydronaphthalenes **6** (Scheme 1, *bottom*), while 1,2-disubstituted alkenes give rise to Heck-type couplings without further cyclization (Scheme 2).

Results and Discussion

Our first attempt focused on the reaction of α-bromochalcone **1a** (employed as an *E/Z*-mixture 11:89) and 5-bromo-1-pentene **4a** in DMF (Table 1). We established that there is no reaction upon irradiation in the absence of a catalyst (entry 1). Electron transfer to **1a** ($E_{ox} = -0.84$ V, see the Supporting Information, all potentials here and in the following are given against SCE) using well established photoredox catalysts for the oxidative quenching cycle, i.e., $Ru(bpy)_3Cl_2$ ($E_{Ru2+*/Ru3+} = -0.81$ V,[15a] bpy = 2,2′-bipyridine, entry 2), or $Cu(dap)_2Cl$ [$E_{Cu+*/Cu2+} = -1.43$ V,[15b] dap = 2,9-bis(4-anisyl)-1,10-phenanthroline, entry 3] did not result in any conversion, suggesting that the process [back electron transfer from **14** (Scheme 5) or subse-



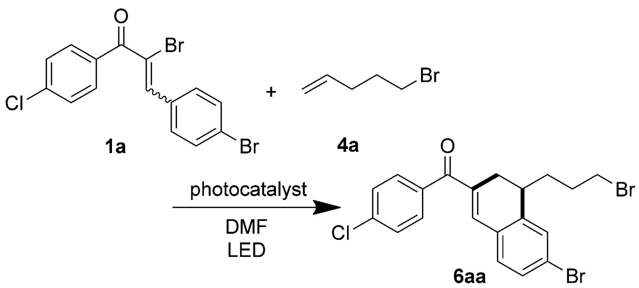
Scheme 1. Coupling of α-bromochalcones **1** with olefins – possible reactions pathways.



Scheme 2. Heck-type coupling and allylation.

quent intermediates] necessary to close the catalytic cycle does not efficiently occur. Turning to iridium catalysts, however, was more promising: Ir[(ppy)$_2$(dtbbpy)]PF$_6$ (E = –0.96 V,[15c] ppy = 2-phenylpyridine, dtbbpy = 4,4-di-*tert*-butyl-2,2-dipyridyl, entry 4) showed the formation of cyclized product **6aa**, however, only in trace amounts. Gratifyingly, other iridium catalysts such as [Ir{dF(CF$_3$)ppy}$_2$(dtbbpy)]PF$_6$ [E = –0.89 V,[15c] dF(CF$_3$)ppy = 2-(2,4-difluorophenyl)-5-trifluoromethylpyridine, dtbbpy = 4,4-di-*tert*-butyl-2,2-dipyridyl, entry 5] and *fac*-Ir(ppy)$_3$ (E = –1.73 V,[15d] ppy = 2-phenylpyridine, entry 6) at 1 mol% gave **6aa** in 68% and 75% yields, respectively, as the sole product. A final control experiment (entry 7) proved that irradiation in combination with *fac*-Ir(ppy)$_3$ is necessary for the reaction to take place. Noteworthy, no interference of the aromatic as well as the aliphatic bromine present in **1a** and **4a** was observed, demonstrating the selective activation of the vinyl bromide, which obviously benefits from the adjacent benzoyl group. Attempts to access the reductive quenching cycle of *fac*-Ir(ppy)$_3$ by reacting **1a** and **4a** in the presence of the sacrificial electron donor triethylamine only resulted in the reductive dehalogenation of **1a** in 82% yield (entry 8).

Table 1. Optimization of the reaction conditions.[a]

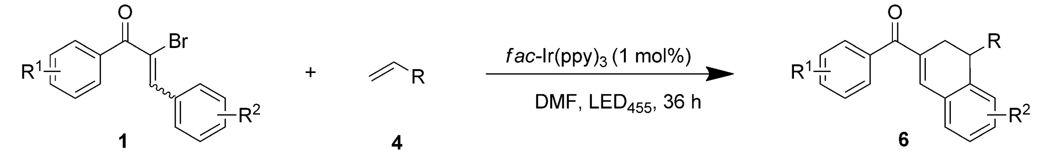


Entry	Photocatalyst	Yield [%][b]
1	no photocatalyst, 455 nm	no reaction
2	Ru(bpy)$_3$Cl$_2$, 455 nm	no reaction
3	Cu(dap)$_2$Cl, 530 nm	no reaction
4	Ir[(ppy)$_2$(dtbbpy)]PF$_6$, 455 nm	traces
5	[Ir{dF(CF$_3$)ppy}$_2$(dtbbpy)]PF$_6$, 420 nm	68
6	***fac*-Ir(ppy)$_3$, 455 nm**	**75**
7	*fac*-Ir(ppy)$_3$, no light	no reaction
8	*fac*-Ir(ppy)$_3$, NEt$_3$, 455 nm	dehalogenation

[a] *Reaction conditions:* **1a** (1 equiv.), **4a** (3 equiv.), photocatalyst (1 mol%).
[b] Isolated yield.

Having identified *fac*-Ir(ppy)$_3$ as the catalyst of choice for the process, we evaluated the substrate scope for this transformation (Table 2). Electron-donating and electron-withdrawing substituents in either ring of the chalcone (**1a–1g**) were amenable in the process, giving the cyclization product **6** in moderate to good yields. However, nitro substituents in either ring (Table 2, entries 21 and 22) are not tolerated; no conversion of the starting materials was observed, albeit electron transfer to **1n**, **1o** judged by their reduction potential (see the Supporting Information for the reduction potentials of α-bromochalcones) should be easily possible by Ir(ppy)$_3$. Since nitro groups are

Table 2. Reaction of α-bromochalcones with olefins.[a]



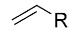
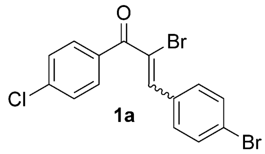
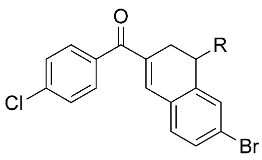
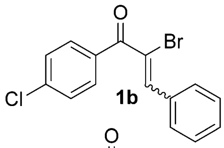
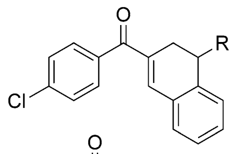
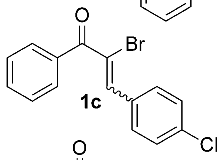
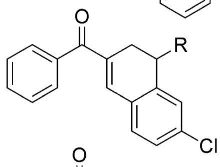
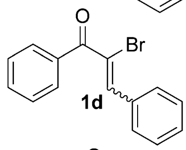
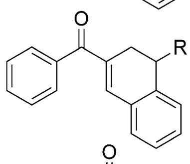
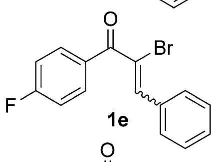
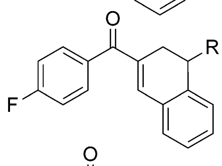
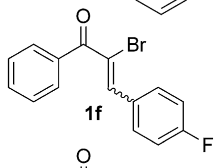
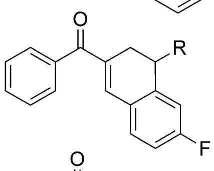
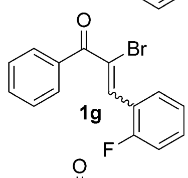
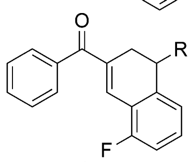
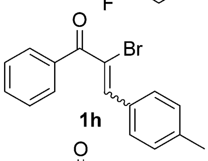
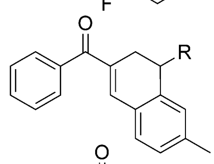
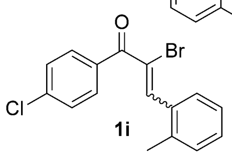
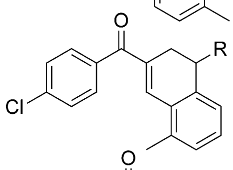
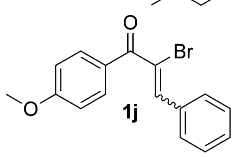
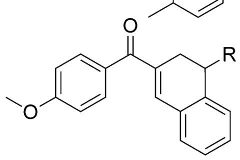
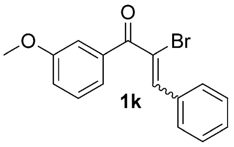
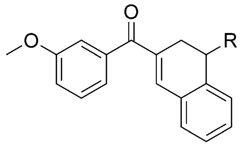
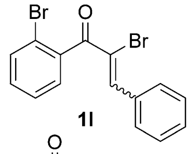
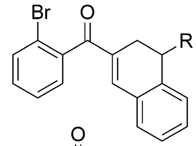
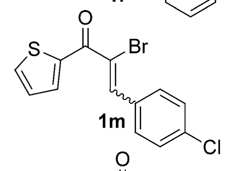
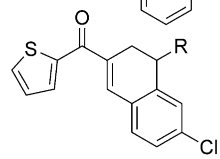
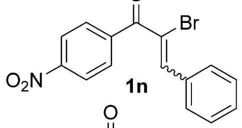
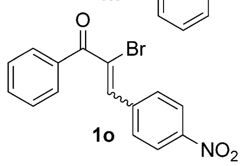
Entry	Substrate (1)	Olefin (2)	Product (6)		Yield [%]
					
1	 1a	**4a**, R = $(CH_2)_3Br$		**6aa**	75
2		**4b**, R = CH_2OAc		**6ab**	78
3		**4c**, R = CH_2SiMe3		**6ac**	70
4		**4d**, R = CH_2Ph		**6ad**	82
5		**4e**, R = CH_2NHBoc		**6ae**	68
6		**4f**, R = $(CH_2)_5CH_3$		**6af**	73
7	 1b	**4a**, R = $(CH_2)_3Br$		**6ba**	73
8		**4g**, R = CH_2Cl		**6bg**	60
9		**4h**, R = CH_2O-2,4-di-Cl-C_6H_3		**6bh**	56
10	 1c	**4a**, R = $(CH_2)_3Br$		**6ca**	80
11	 1d	**4a**, R = $(CH_2)_3Br$		**6da**	71
12	 1e	**4i**, R = $(CH_2)_2Br$		**6ei**	62
13	 1f	**4a**, R = $(CH_2)_3Br$		**6fa**	56
14	 1g	**4a**, R = $(CH_2)_3Br$		**6ga**	51
15	 1h	**4c**, R = CH_2SiMe_3		**6hc**	52
16	 1i	**4a**, R = $(CH_2)_3Br$		**6ia**	41
17	 1j	**4c**, R = CH_2SiMe_3		**6jc**	47

Table 2. (Continued)

Entry	Substrate (1)	Olefin (2)	Product (6)		Yield [%]
18	 **1k**	**4c**, R = CH_2SiMe_3		**6kc**	59
19	 **1l**	**4a**, R = $(CH_2)_3Br$		**6la**	67
20	 **1m**	**4c**, R = CH_2SiMe_3		**6mc**	60
21	 **1n**	**4a**, R = $(CH_2)_3Br$	—	—	
22	 **1o**	**4a**, R = $(CH_2)_3Br$	—	—	

[a] *Reaction conditions:* **1** (0.5 mmol), **4** (3 equiv,) and photocatalyst (1 mol%) in dry DMF (2.0 mL) was irradiated for 36 h with an LED light source (455 nm).
[b] Yield of isolated product.

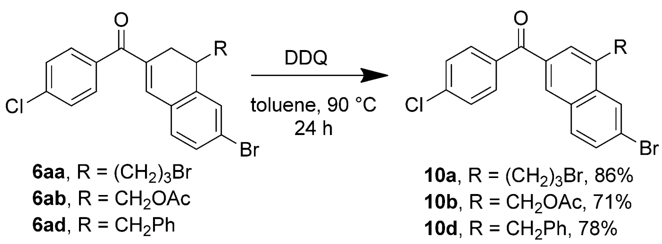
strong electron acceptors themselves, we assume that reversible electron transfer takes place from the photocatalyst to the nitroarene ring, thus preventing the activation of vinyl bromide. Replacement of the aryl substituent next to the carbonyl group by a heteroarene was possible (entry 20), but we were unable to replace the aryl group by non-arene substituents. Apparently, the mesomeric stabilization of the former is necessary for a vinyl radical to form or to react further.

A broad range of terminal alkenes **2a–h** could be employed in the process, tolerating functional groups (Br, Cl, OAc, 2,4-di-ClC_6H_3O, $SiMe_3$, Bn, NHBoc) well. In all cases the products were formed cleanly without by-products, the moderate yields for some entries were due to incomplete conversion of the starting materials.

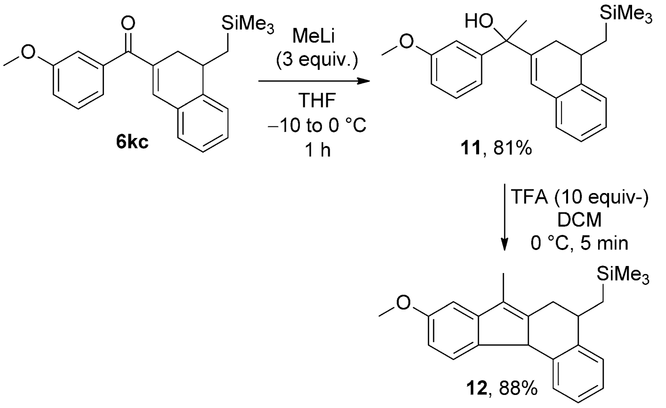
Cyclohexene, representing a 1,1-disubstituted alkene, was also a suitable coupling partner, but followed a different reaction pathway. In this case, the second C–C bond formation did not take place, thus forming the Heck coupling-type product **7** (Scheme 2).[16] For allyl bromides **8**, bromide elimination was observed leading to allylated chalcones **9** as an *E*/*Z* mixture (Scheme 2), with no cyclization product being observed. Obviously, good radical leaving groups terminate the process by elimination effectively.

The resulting dihydronaphthalenes **6**, besides their relevance as building blocks themselves,[1–4] can be easily oxidized with DDQ to substituted naphthalenes **10** (Scheme 3).[17]

Noteworthy, we did not observe the formation of naphthalenes **10** already in the photocyclization, although iridium catalysts are known to act as hydrogenation/dehydrogenation catalysts, and such aromatization reactions have been described as a follow-up reaction in related processes.[14,18]



Scheme 3. Conversion of dihydronaphthalenes to naphthalenes.

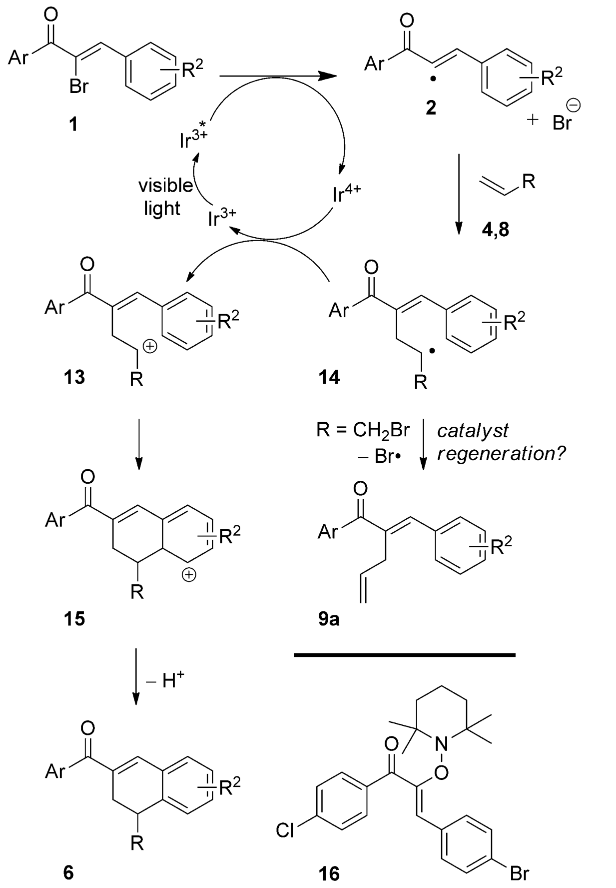


Scheme 4. Synthesis of 5*H*-benzo[c]fluorene derivative **12**.

To further illustrate the utility of the dihydronaphthalene products, **6kc** was converted to the 5*H*-benzo[c]fluorene derivative **12** in high yield by subjecting it to methyllithium addition to form the tertiary alcohol **11**, followed by TFA-mediated cyclization (Scheme 4).[19]

A plausible reaction mechanism for photocoupling between **1** and **4**, being in line with related studies,[16,20] is depicted in Scheme 5. Oxidative quenching of excited $^{*}Ir^{3+}$ by α-bromochalcone **1** results in the formation of vinyl radical **2**, which was detected by mass spectrometry as trapping adduct **16** in an independent experiment between **1a** and TEMPO.

The radical **2** then adds to the olefin **4** or **8** to form the radical **14** as a key intermediate, which subsequently can undergo different reaction pathways depending on the substitution pattern of the alkene. Terminal alkenes **4** undergo back electron transfer to form cation **13** and thus regenerating the catalyst. Electrophilic ring closure to **15** and proton elimination gives rise to dihydronaphthalene **6**. For the sterically more hindered cyclohexene (*cf.* Scheme 2) the reaction terminates by a proton loss yielding **7** without subsequent cyclization, presumably due to a lack of proper orientation of the cyclohexyl moiety for cyclization caused by unfavorable 1,3-interactions between the cyclohexane ring and the aryl substituent located at the carbonyl group. For allylic bromides **8**, instead of the cyclization the elimination of a bromine radical (for X = Br) is observed to form allylated chalcones **9**, which opens up the question how the catalyst is regenerated, i.e., reduction of Ir^{4+} to Ir^{3+} is necessary. The termination of photoredox catalzyed allylations *via* elimination of tosyl radicals has been recently reported,[20] in which a tertiary amine is present that acts as a sacrificial electron donor to regenerate the photocatalyst. Under our reaction conditions, the presence of dimethylamine, being generated by decomposition of the solvent DMF due to the generation of HBr in the course of the reaction, could take the role as electron donor.



Scheme 5. Proposed reaction mechanism for the photoredox catalyzed C–C coupling between chalcones and olefins.

Conclusions

In conclusion, a photochemical synthesis of dihydronaphthalenes **6** has been achieved by intermolecular vinyl radical anellations to olefins utilizing the iridium catalyst *fac*-Ir(ppy)$_3$ and visible light. The dihydronaphthalene products can be converted to substituted naphthalenes and furthermore to polycyclic indenes. Mechanistic investigations support the generation of vinyl radicals as key intermediates, but further investigations on the overall catalytic cycle for the observed allylations are necessary to fully explain the experimental results obtained.

Experimental Section

Typical Procedure for the Photoredox Catalyzed Cascade Cyclization

An oven-dried 15-mL Schlenk tube equipped with a plastic septum and magnetic stir bar was charged with Ir(ppy)$_3$ (1 mol%), α-bromochalcone **1a** (0.5 mmol, 1.0 equiv.) and the olefin **2a** (1.5 mmol, 3.0 equiv.). The flask was purged with a stream of nitrogen and 2.0 mL of dry dimethylformamide were added. The resultant mixture was degassed by a freeze-pump-thaw procedure (3 cycles). The tube was sealed with an internal irradiation set up (an LED stick inside) and irradiated for 36 h. After the completion of the reaction (as judged by TLC analysis), the mixture was transferred to a separating funnel, diluted with 15 mL of ethyl acetate and washed with 20 mL of water. The aqueous layer was washed with ethyl acetate (3×10 mL) and the combined organic layer was dried over anhydrous sodium sulfate, solvent was removed under vacuum and the residue was subjected to colum chromatography on silica gel, using PE/EA as solvent system to furnush the pure **6aa** as a white solid; yield: 176 mg (75%). R_f (EtOAc/hexane 1:9): 0.45; mp 161–163 °C; IR (neat): ν = 3027, 1652, 1583, 1276, 1162, 732, 637 cm^{-1}; ^1H NMR (300 MHz, $CDCl_3$): δ = 7.71–7.63 (m, 2H), 7.49–7.42 (m, 2H), 7.40–7.33 (m, 2H), 7.03 (dd, J = 5.3, 3.2 Hz, 2H), 3.47–3.32 (m, 2H), 2.95 (qd, J = 6.9, 4.2 Hz, 1H), 2.85 (dd, J = 17.2, 4.0 Hz, 1H), 2.72 (ddd, J = 17.2, 6.7 Hz, 2.3, 1H), 2.05–1.63 (m, 4H); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 195.84, 142.77, 138.44, 138.28, 136.28, 135.77, 130.79, 130.66, 130.55, 130.48, 130.29, 128.72, 124.17, 36.59, 33.62, 32.48, 30.15, 27.38; HR-MS (ESI): m/z = 466.9417, calcd. for $C_{20}H_{18}Br_2ClO$ $[M+H]^+$: 466.9413.

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